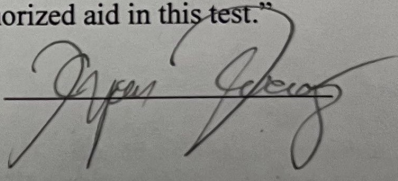


Midterm Exam
MAE C163A/C263A Fall 2022Name: Ryan Rossmango
UID: 705 377 683

- Make sure you write your name and ID number.
- You can use a laptop.
- The exam is open book and notes.
- To receive credit, please be **neat** and show all key steps of your solutions clearly.
- Don't forget to submit it on **Gradescope**.
- Please sign the following pledge before beginning the exam:

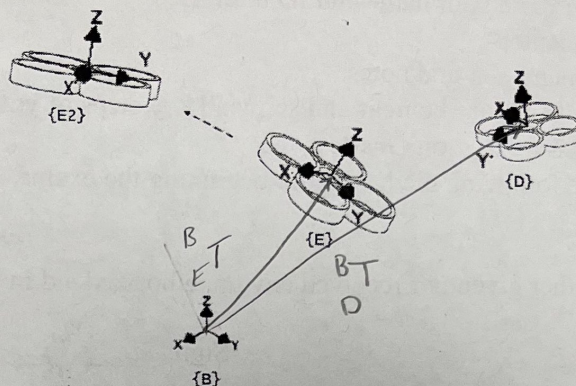
"I have neither given nor received any unauthorized aid in this test."

Sign: 

Problem #	Score
1	/20
2	/30
3	/20
4	/30
Total	/100

Problem 1 (20 pts)

A laboratory is testing quadcopters. With markers attached to each quadcopter, the lab's motion capture system records the position and orientation of the quadcopters with respect to the fixed base frame $\{B\}$. (Figure does not show the actual scale or orientation)



Quadcopters D and E are hovering. Their positions and orientations with respect to frame $\{B\}$ are given in the transformation matrices below. (All units are in meters and degrees)

$${}^B_D T = \begin{bmatrix} 1 & 0 & 0 & -20 \\ 0 & c(30^\circ) & -s(30^\circ) & -20 \\ 0 & s(30^\circ) & c(30^\circ) & 80 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad {}^B_E T = \begin{bmatrix} c(30^\circ) & -s(30^\circ) & 0 & 10 \\ s(30^\circ) & c(30^\circ) & 0 & 50 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$* c(30^\circ) = \cos(30^\circ) \approx 0.866, s(30^\circ) = \sin(30^\circ) = 0.5$$

- 1) Find the position & orientation of copter E with respect to copter D (${}^D_E T$) as well as the position & orientation of copter D with respect to copter E (${}^E_D T$). (10 pts)

$${}^D_E T = {}^D_B T {}^B_E T = {}^B_D T^{-1} {}^B_E T = \text{MATLAB}$$

$${}^D_E T = \begin{pmatrix} 0.866 & -0.5 & 0 & 30 \\ 0.433 & 0.75 & 0.5 & 20.6218 \\ -0.25 & -0.433 & 0.866 & -104.282 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$${}^E_D T = {}^D_E T^{-1} = \text{MATLAB}$$

$${}^E_D T = \begin{pmatrix} 0.866 & 0.433 & -0.25 & -60.9808 \\ -0.5 & 0.75 & -0.433 & -45.6218 \\ 0 & 0.5 & 0.866 & 80 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

- 2) Copter E (frame $\{E\}$) rotates +30 degrees about X_E axis and has a translation of $\begin{bmatrix} 20 \\ -10 \\ 0 \end{bmatrix}$ with

respect to frame $\{D\}$. Thus, copter E (frame $\{E\}$) will have a new location and orientation, we call this new frame $\{E_2\}$. Find the new orientation and position of copter E (frame $\{E_2\}$) with respect to the base frame $\{B\}$ (B_T). (10 pts; hint: find D_T first)

$${}^E_R = R_{X_E} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & c30 & -s30 \\ 0 & s30 & c30 \end{pmatrix} \rightarrow \text{want } {}^D_R = {}^D_T {}^E_R, \text{ in order to make the transformation } {}^D_T$$

Need 4x4 size, so $\begin{pmatrix} {}^D_R \\ {}^E_R \\ 0 & 0 & 0 \end{pmatrix}_{4 \times 3} = {}^D_T \begin{pmatrix} {}^E_R \\ {}^E_R \\ 0 & 0 & 0 \end{pmatrix}_{4 \times 3}$ so ${}^D_R = \begin{pmatrix} 0.866 & -0.433 & 0.25 \\ 0.433 & 0.8995 & 0.058 \\ -0.25 & 0.058 & 0.9665 \end{pmatrix}$

To make D_T , we have D_R and ${}^D_{\text{Translation}}$

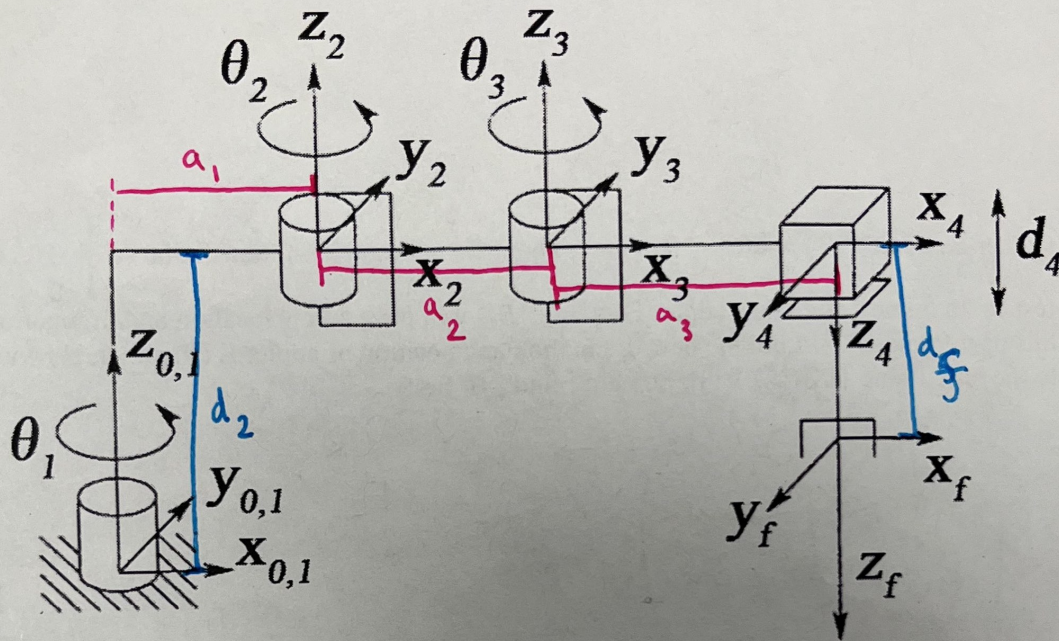
$$\hookrightarrow = \left(\begin{array}{c|c} {}^D_R & {}^D_{\text{Translation}} \\ \hline {}^E_R & \\ 0 & 0 & 0 & 1 \end{array} \right) = \begin{pmatrix} 0.866 & -0.433 & 0.25 & 20 \\ 0.433 & 0.8995 & 0.058 & -10 \\ -0.25 & 0.058 & 0.9665 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$${}^B_T = {}^B_T {}^D_T {}^D_T {}^E_T$$

$${}^B_T = \begin{pmatrix} 0.866 & -0.433 & 0.25 & 0 \\ 0.5 & 0.75 & -0.433 & -28.6603 \\ 0 & 0.5 & 0.866 & 75 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Problem 2 (30 pts)

Consider the robotic arm shown below. It has 3 revolute joints and 1 prismatic joint. The coordinate frames are given below. Note that frame $\{f\}$ is the end-effector frame.



- 1) Complete the DH table below.
Also, **draw and label** any a_{i-1} and d_i distances on the figure above. (10 pts)

$i-1$	i	α_{i-1}	a_{i-1}	d_i	θ_i
0	1	0	0	0	θ_1
1	2	0	a_1	d_2	θ_2
2	3	0	a_2	0	θ_3
3	4	$+180^\circ$	a_3	d_4	0
4	f	0	0	d_f	0

x_2 and x_3 are same axis

$$\begin{aligned}
 x &= \theta_1 & y &= \theta_2 & a_1 &= a & d_2 &= d \\
 z &= \theta_3 & a_2 &= b & a_3 &= c & d_4 &= e & d_f &= f
 \end{aligned}$$

2) Find the individual transformation matrices below. (15 pts)

$${}^0_1T = \begin{bmatrix} c_1 & -s_1 & 0 & 0 \\ s_1 & c_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^1_2T = \begin{bmatrix} c_2 & -s_2 & 0 & a_1 \\ s_2 & c_2 & 0 & 0 \\ 0 & 0 & 1 & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^2_3T = \begin{bmatrix} c_3 & -s_3 & 0 & a_2 \\ s_3 & c_3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^3_4T = \begin{bmatrix} 1 & 0 & 0 & a_3 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & -d_4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^4_fT = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_f \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

3) Find the overall simplified transformation matrix 0_fT . (5 pts)

$${}^0_fT = {}^0_1T {}^1_2T {}^2_3T {}^3_4T {}^4_fT$$

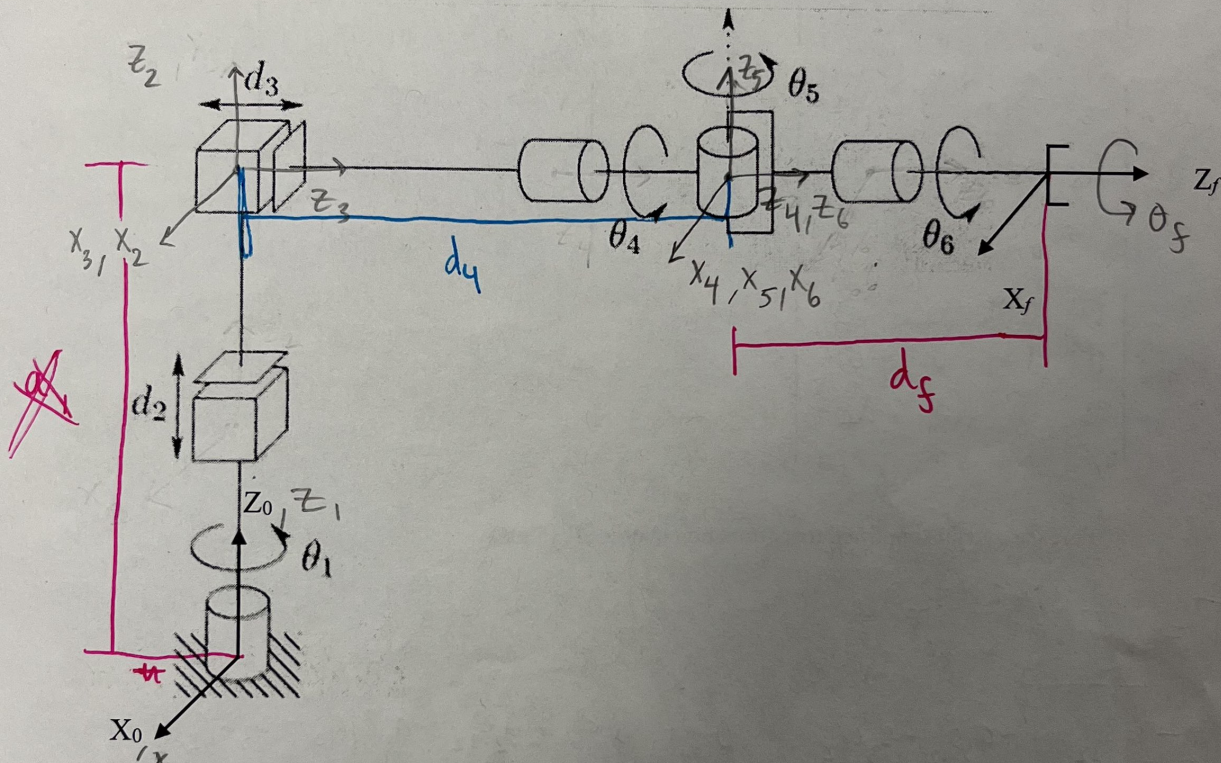
$$\begin{aligned}
 &= \begin{pmatrix} c_3 c_{12} - s_3 s_{12} & +c_3 s_{12} - s_3 c_{12} & 0 & a_1 c_1 + a_2 c_{12} + a_3 c_3 c_{12} - s_3 s_{12} \\ c_3 s_{12} + s_3 c_{12} & -c_3 c_{12} - s_3 s_{12} & 0 & a_1 s_1 + a_2 s_{12} + a_3 c_3 s_{12} + s_3 c_{12} \\ 0 & 0 & -1 & -d_f + d_2 - d_4 \\ 0 & 0 & 0 & 1 \end{pmatrix}
 \end{aligned}$$

used
MATLAB
calculator
called
Symbolab
online

Problem 3 (20 pts)

Consider the robotic arm shown below. Note that frame $\{f\}$ is the end-effector frame.

- 1) Draw the remaining coordinate frames (z and x-axis only) on the figure below. (10 pts)

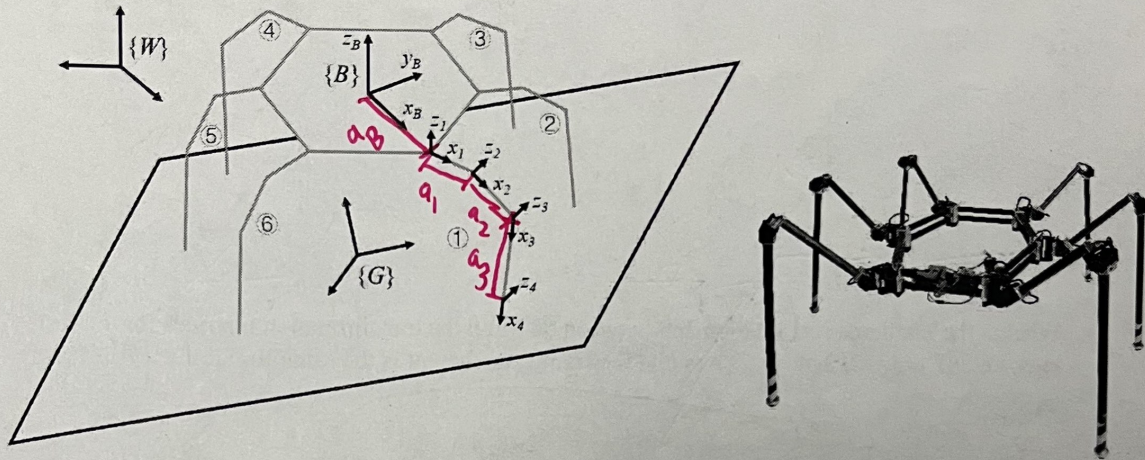


- 2) Complete the DH table below.
Also, draw and label any a_{i-1} and d_i distances on the figure above. (10 pts)

$i-1$	i	a_{i-1}	a_{i-1}	d_i	θ_i
0	1	0	0	0	θ_1
1	2	0	0	d_2	0
2	3	-90	0	d_3	0
3	4	0	0	d_4	θ_4
4	5	+90	0	0	θ_5
5	6	-90	0	0	θ_6
6	f	0	0	d_7	θ_f

Problem 4 (30 pts)

Consider a hexapod robot shown below. It has 6 identical legs and each leg has 3 revolute joints. Note that $\{W\}$, $\{B\}$, and $\{G\}$ are the world frame, body frame, and ground frame, respectively.



- 1) Complete the DH table below for leg 1. Note that frame $\{4\}$ is the end-effector/toe frame; frame $\{B\}$ is considered as the base frame; z_B -axis is perpendicular to the regular hexagon plane; x_B -axis is pointing to the origin of frame $\{1\}$; axes z_2, z_3, z_4 are parallel to each other. Also, **draw and label** any a_{i-1} and d_i distances on the left figure above. (10 pts)

$i-1$	i	α_{i-1}	a_{i-1}	d_i	θ_i
B	1	0	a_B	0	θ_1
1	2	-90	a_1	0	θ_2
2	3	0	a_2	0	θ_3
3	4	0	a_3	0	θ_4

- 2) Find the overall simplified transformation matrix B_4T_1 , where the index 1 denotes leg 1. (10 pts)

$${}^B_4T = {}^B_1T {}^1_2T {}^2_3T {}^3_4T$$

Don't have MATLAB
symbolics tool box.

- 3) What is the mathematical relationship between B_4T_1 and the transformation matrices for the rest legs, i.e., B_4T_j , $j = 2, 3, 4, 5, 6$? Assume the frame attachment is the same for all the legs. (5 pts)

$${}^B_4T_1 = {}^B_4T_j \cdot {}^B_4T_j$$

- 4) If we know the toe position of leg 6 in $\{W\}$, denoted as Wt_6 , as well as W_BT , which describes frame $\{B\}$ with respect to frame $\{W\}$, how can we determine this toe position in $\{B\}$? (5 pts)

$${}^Wt_6 = {}^W_BT \cdot {}^Bt_6 \quad \leftarrow \text{general form}$$

$$\begin{matrix} {}^Bt_6 & = & {}^B_WT & {}^Wt_6 \\ \underbrace{\phantom{{}^Bt_6}}_{3 \times 1} & & \underbrace{\phantom{{}^B_WT}}_{4 \times 4} & \underbrace{\phantom{{}^Wt_6}}_{3 \times 1} \end{matrix}$$

$$\begin{pmatrix} {}^Bt_6 \\ 1 \end{pmatrix} = {}^W_BT^{-1} \begin{pmatrix} {}^Wt_6 \\ 1 \end{pmatrix}$$

$$\rightarrow \begin{pmatrix} {}^Bt_6 \\ 1 \end{pmatrix} = {}^W_BT \begin{pmatrix} {}^Wt_6 \\ 1 \end{pmatrix}$$

4x1 4x4 4x1 ✓

Do this calculation
to get Bt_6